Understanding Haptic Authoring Tools through a Taxonomy and Descriptive Analysis

Juliette Regimbal
Department of Electrical and Computer Engineering
McGill University
Montréal, Quebec, Canada
juliette.regimbal@mail.mcgill.ca

Cyan Kuo
Department of Electrical and Computer Engineering
McGill University
Montreal, Quebec, Canada
cyan.kuo@mail.mcgill.ca

Jeremy R Cooperstock
Centre for Intelligent Machines
McGill University
Montreal, Quebec, Canada
jer@cim.mcgill.ca

Abstract

Well-designed haptic interactions can improve user experience, but implementing them is often challenging. Haptic authoring tools help alleviate this difficulty, but they are suited for use with different hardware, applications, and practitioners. As such, novices to haptic design need to determine which tools, if any, are suitable for their projects. Similarly, practitioners improving authoring tools and extending them to new contexts need to understand advancements made in prior work to identify opportunities for improvement. Unfortunately, the haptic authoring literature is disorganized, hindering practitioners from succeeding in these tasks. To remedy these shortcomings, we developed a taxonomy of haptic authoring tools, used it to identify gaps in the literature, and analyzed trends in the development of the tools. We also present the systematic literature review and study with thirteen haptics practitioners that were used to produce these results, then discuss what they mean for future research in haptic authoring.

CCS Concepts

• **Human-centered computing → Haptic devices**.

Keywords

Authoring tools, haptic design, taxonomy

ACM Reference Format:
Juliette Regimbal, Cyan Kuo, and Jeremy R Cooperstock. 2026. Understanding Haptic Authoring Tools through a Taxonomy and Descriptive Analysis. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain.* ACM, New York, NY, USA, 17 pages. https://doi.org/10.1145/3772318.3790575

1 Introduction

Haptic interactions leverage the sense of touch to make computer interfaces more useful and enjoyable for users [21]. However, creating haptic effects is not an easy task. Haptic design and implementation require considerable skill and experience [81, 87], preventing many practitioners from including haptics in their applications. Consequently, numerous authoring tools for haptic effects have been developed that support an assortment of devices [27, 37, 102] and are intended for different domains of work [49, 65, 95].

This work is licensed under a Creative Commons Attribution 4.0 International License.
CHI '26, Barcelona, Spain
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2278-3/26/04
https://doi.org/10.1145/3772318.3790575

While these tools make it easier than ever for people to begin creating haptic effects and interactions, there remain significant gaps in the support available. The sense of touch encompasses vastly different stimuli, such as forces, vibration, skin stretch, and temperature changes [61, 98]. A variety of hardware devices are needed to create these different types of haptic effects. For example, the devices used to create vibration effects vary in terms of actuator type, number, and how they are deployed to transfer energy to the human body [14]. Additionally, practitioners have context-specific requirements for authoring tools since the characteristics of haptic effects differ depending on their intended purpose. A musician, for example, may want to create physically realistic haptic feedback for an instrument [10, 54], while a video game designer may have no such desire [92].

Unsurprisingly, no one tool can do it all. Thus, it is important for practitioners interested in haptic authoring to understand what has already been developed and studied. For many, this interest is practical: they need to know what they can use in their own projects. For those researching and developing authoring tools, they instead need to find inspiration and opportunities in past work in order to properly guide their future efforts. Unfortunately, neither group is currently well supported for their task. The literature on haptic authoring is scattered, requiring considerable time and effort to identify what previous work exists. As a result, practitioners often focus only on what they already know, potentially missing tools they could use or insights that could benefit their work.

To address this situation, we seek to create a collection of haptic authoring tools, categorize them using a novel taxonomy refined through a user study, and conduct a descriptive analysis of them to guide other practitioners. This process includes six stages:

(1) An initial literature review of haptic authoring tools intended to form a high-level understanding of the topic.
(2) The creation of an initial taxonomy based on previous literature and open coding of the tools.
(3) A study with hapticians to evaluate the taxonomy and identify characteristics of haptic authoring tools relevant to their work.
(4) A systematic review of the literature based on the results of this study to form a final collection of 79 authoring tools relevant to the work of haptics practitioners.
(5) Revising the taxonomy into its current form, which consists of 19 different fields.
(6) Conducting a descriptive analysis of the collection through the lens of the taxonomy to identify trends relevant to practitioners.

In this context, we define as a haptic authoring tool any software or toolkit whose documentation or publications indicate it was *intended* to be used to create haptic effects. The decision to focus on intent, which excludes general purpose programming languages, audio editing tools, and other software repurposed by haptics practitioners [81], was made for practicality. Including every software that could possibly be adopted to create stimuli perceived as touch would be infeasible. A definition based on intent as expressed through publicly accessible records provides reasonably objective criteria for inclusion while not excluding software on the basis of its complexity, the hardware it supports, or its user interface.

We evaluate the current taxonomy based on its descriptive, evaluative, and generative power; three criteria established by Beaudouin-Lafon [4] and used to assess other taxonomies, notably the one for grounded force-feedback devices in Haptipedia [88]. The evaluative power of our taxonomy is demonstrated via a study where it is used by haptics practitioners to select suitable authoring tools for different design scenarios (Section 6). Its descriptive power is demonstrated via a descriptive analysis of the 79 collected tools (Section 7). Finally, we demonstrate its generative power by identifying gaps in the haptic authoring tool literature based on the results of this descriptive analysis (Section 8).

The contributions of this work are the taxonomy and the descriptive analysis. An interactive website to navigate the collection of the tools annotated using the taxonomy is available at https://srl.mcgill.ca/hat-box to help readers apply our results to their own projects. For offline use, a CSV file listing the tools and their annotations is included in the supplemental materials.

2 Background

2.1 Role of haptic authoring tools

Modern haptic authoring tools have roots in the early 2000s, with projects such as Skinscape [34], GENESIS [11], the Hapticon Editor [28], and a VRML-based system created by O'Malley and Hughes [70]. Though developed in different contexts and for very different devices, all served to simplify the design of haptic content. Authoring tools users include musicians [11, 34], creators of haptically augmented virtual reality (VR) content [70], and professionals already working with haptics [28].

For those unfamiliar with the technology, the added value is clear: authoring tools are an entry point to working with haptics. Seifi et al. reported that novice haptic designers need assistance in learning to use new devices, to communicate with their peers, and to apply accepted design practices when creating haptic content and situating it within a multisensory context. Haptic authoring tools were suggested as a means of providing scaffolding for novices as they gain expertise [87], a principle put into practice in projects such as VibScoreEditor [52], the TECHTILE toolkit [63], FeelCraft [82], and HITPROTO [75, 76].

Hapticians continue to experience challenges even as they gain expertise. Collaboration and design iteration, for instance, were identified as difficult tasks by Schneider et al. in interviews with haptics practitioners from industry and academia [81]. Features such as a shared library of effects [106], multiple actuators to enable collaborators to feel an effect simultaneously [84], or a human-readable storage format [86] can be used to support haptic design work within a team. Efficiency also is a factor. The ability to rapidly prototype complex interactions [43, 65, 76] or run existing media on new devices [16, 65, 74] can considerably increase the productivity of hapticians.

2.2 Efforts to categorize haptic authoring tools

To the knowledge of the authors, no systematic review of haptic authoring as a whole has yet been undertaken, but several articles have described categories and guidelines for authoring tools in particular contexts. In a discussion of tactile authoring and prototyping prior to presenting TactiPEd, Panëels et al. described a historical trend in which authoring tools shifted from providing low-level control over hardware to presenting more abstract metaphors to users [74]. They also identified clusters of tools based on the type of metaphor used (touch, visual, or audio), the number of supported actuators, and whether the spatial layout of the actuators was considered.

Ryu and Choi compared the posVibEditor with Immersion Corporation's VibeTonz and the Haptic Icon Prototyper [96], focusing mainly on available features, such as pattern designs and multiple actuators, and the presence of interface elements, such as a multi-channel timeline or drag-and-drop controls [80]. They noted the ability to play back haptic effects in all three editors. Cuartielles et al. described the importance of direct feedback and the need for editors of full body effects to provide an interface that represents the whole body [18].

Choi and Kuchenbecker described tactile authoring in multimedia contexts as ranging on a scale from fully manual creation to fully automatic translation from pre-existing media [14]. Examples of either extreme are mentioned alongside the possible design of software that provides a combination of direct control with automated features.

Frisson et al. formulated two categories of force-feedback haptic frameworks. The first uses physical modelling to produce complex, realistic audio-haptic scenes. The second "signal modelling" group creates haptic stimuli using features that are inspired by audio tools, but often without producing sound [31].

Guidelines for authoring tools have been proposed by some authors through literature review and their own experiences as tool creators. Schneider et al. developed design requirements from literature and expert interviews in order to create Mango [83]. While some requirements are specific to tactile animation, the purpose for which the tool was developed, most of them can be applied broadly. These general requirements include real-time playback, configuration to support different hardware devices, libraries of effects, and the visualization of haptic effects using another sensory modality [83].

Similarly, van Oosterhout et al. developed a summary of guidelines of authoring tools from prior work that recommends support for real-time playback, an iterative design process, a direct representation of how effects will be displayed on the device, and a high degree of flexibility and interoperability. They indicated that

their guidelines would need to be expanded to respond to the requirements of a particular use case, referencing those for tactile animation developed for Mango [100].

Seifi et al. developed a set of requirements for Feellustrator, a tool for ultrasound mid-air haptics, by interviewing ultrasound designers [86]. These requirements focused on supporting a high level of control over the design of the stimuli and how they would be rendered, accessing basic patterns in a library, creating complex patterns out of simpler ones, and previewing the designed effect with and without hardware connected. The eleven "essential" features indicated by the interviewees were included in the final design of Feellustrator.

These efforts to describe haptic authoring tools and guide their development provide useful information to practitioners. However, the focus has remained on tools for specific applications or devices. A project with a broader scope, such as the one described in this paper, may seek to organize information about authoring tools in a structured way, such as by using a taxonomy.

2.3 Taxonomies in HCI and haptics

Taxonomies have been used in HCI [51, 57] and haptics [72, 88, 89, 103] to describe differences between the prior work on a topic and to aid in manipulating and visualizing relevant information [88]. They are typically created iteratively with the taxonomic labels being determined through systematic literature reviews [7, 57, 72, 88, 103] and user studies [88, 89]. Beaudouin-Lafon described three properties of a useful taxonomy:

- it can **describe** what exists;
- it can help **evaluate** alternative options; and
- it can help **generate** new ideas [4].

These criteria align with Wobbrock and Kientz's definition of an HCI survey contribution [107]. Frequently, authors present the results of their taxonomy through an interactive website to aid readers in applying the results to their own work [7, 88, 89].

Following the example set by these previous publications, we conduct a systematic review of haptic authoring tools and an evaluative study with haptics practitioners to develop our taxonomy and inform our analysis of the literature. We also create a website to make these results more accessible to readers and to facilitate updates to our collection and taxonomy as research in haptic authoring continues.

3 Initial Taxonomy

Our intention in developing a taxonomy of haptic authoring tools was to help haptics practitioners in two tasks: finding resources that meet their needs and creating novel authoring tools. To begin our work towards these goals, we conducted an informal review of haptic authoring tools and analyzed the results to develop an initial version of the taxonomy.

An initial set of 51 tools was identified through discussions with haptics practitioners, searches of online databases, and following references within papers. Thirty-nine of these tools were published in academic journals or conferences, two were the subject of master's theses, and ten were commercial products included at the suggestion of haptics practitioners with whom we spoke. Through the practitioner conversations, we encountered differing ideas of what haptic authoring tools were, leading to the construction of the definition for haptic authoring tools given in Section 1 of "any software or toolkit intended to be used to create haptic effects."

Taxonomy creation began with the open coding of the original 51 tools by the first author. Since the hardware controlled by the tools is vastly different, and many tools could not be obtained for in-depth testing, the author did not attempt to compare the quality or realism of haptic effects that could be produced by each one. Instead, the author focused on how users interact with the tools to create and modify effects. When possible, academic papers describing the tool, its documentation, and the tool itself were reviewed. Codes were iteratively defined and revised into candidate labels for the taxonomy through discussions with colleagues working in haptics. Permitted values for each label were developed through the same process with the goal of capturing high-level differences between tools while retaining enough information to aid practitioners in finding resources relevant to their needs.

Twelve labels were developed for the initial taxonomy and grouped into three categories: hardware, user interaction, and metadata. Table 1 shows the labels of this taxonomy alongside their definitions, permitted values, and number of values each tool may have. For example, since a tool may run on multiple platforms, each one can be recorded under the *Platform* label. However, a tool is only released once, thus *Release Year* is restricted to exactly one value. An overview of the categories of the taxonomy is presented in the following paragraphs.

Metadata. *Release Year* provides chronological context for the tools. The *Platform*, *Availability*, and *License* attributes describe practical information to practitioners interested in using a tool. *Availability* and *Platform* describe if a tool can be obtained, at present and by general practitioners, and what software is needed to run it. Additionally, when haptic authoring tools are built upon other *non-haptic* tools, the latter are noted under *Platform*. For example, the platform for the haptic authoring tool Tactile Brush [40] is the audio synthesis program Pure Data. *License* indicates the legal terms under which a practitioner may use and modify the tool.

Hardware. The *Category* attribute describes the type of haptic feedback supported by a tool, such as vibrotactile or force-feedback. *Abstraction* describes how tightly coupled a tool is to its hardware. Its value indicates if an entire class of devices is supported (e.g., voice coil actuators), if the hardware is a consumer product (e.g., a Novint Falcon), or if only custom-built hardware can be used. These attributes are intended to help practitioners find authoring tools capable of supporting particular types of hardware.

Interaction. The remaining attributes pertain to the user interface of the tool and how it can be used to create haptic effects. *Non-Haptic Media* indicates if effects for another sensory modality can be viewed or edited within the tool. The *Iterative Playback* attribute records if users can test and refine effects without leaving the tool, a feature identified as important in prior work [83, 86, 100]. *Driving Feature* broadly indicates if time or in-application actions can be used to trigger or change effects at runtime. *Effect Localization* indicates how a tool supports spatially distributed haptic stimuli. *Design Approaches* identifies the methods by which haptic effects can be created and *UI Metaphors* shows how this functionality is made interactive to the user of the tool. Definitions for the

Table 1: Overview of the initial taxonomy.

	Attribute	Definition	Values	Values per Tool
Metadata	Release Year	The year the tool was released or described in a publication.	Any year	1
	Availability	Whether the tool is available to the public.	Available | Unavailable	1
	License	The terms under which the tool is/was available, if known.	Open Source | Proprietary | Non-Commercial | Unknown	1
	Platform	The software platform(s) that run the tool.	Any name	1 or more
Hardware	Category	The type of hardware device(s).	Vibrotactile | Force-feedback | Temperature	1 or more
	Abstraction	Whether support extends to more specific or generic categories of device.	Custom | Consumer | Class	1
Interaction	Driving Feature	Whether effects change due to time or miscellaneous actions.	Time | Action	1 or more
	Effect Localization	The extent to which effects made with the tool are mapped to an actuator or a separate target.	Device-centric | Location-aware | Target-centric	1
	Iterative Playback	Whether effects can be rendered within the tool, if applicable.	Yes | No | N/A	1
	Non-Haptic Media	Whether non-haptic media is supported in any way.	Audio | Visual	0 or more
	Design Approaches	The method(s) available to create an effect.	Direct | Procedural | Library | Additive | Description	1 or more
	UI Metaphors	Common patterns which define user interaction with the tool.	Track | Keyframe | Music Notation | Dataflow | Demonstration	0 or more

values of *Effect Localization*, *Design Approaches*, and *UI Metaphors* are shown in Table 2.

The tools in the original collection were categorized according to this taxonomy. The results of this effort were encapsulated in a website to aid in navigating and manipulating the data since doing so in a table was cumbersome. The website was then used by participants in an experiment to evaluate and refine the taxonomy.

4 User Evaluation

We conducted a two-activity study[1] to assess and improve the taxonomy's *evaluative* capacity [4]. Thirteen participants with experience building or using haptic authoring tools were recruited through haptics-related mailing lists and word-of-mouth. Prior experience with haptic authoring tools was required so that participants could leverage their knowledge to provide actionable feedback to improve the taxonomy. We sought different levels of haptics experience, from newcomers to seasoned professionals, to benefit from a variety of perspectives. Informed consent was obtained from all participants prior to the start of each activity. Ethics approval was obtained from the McGill University Research Ethics Board under file 22-08-090.

First, all thirteen participants (Table 3) described their prior experience with haptic design. Then, they were shown three different haptic design scenarios where an authoring tool would be useful. Participants used the aforementioned website with the taxonomy-annotated authoring tool collection to select which tool they would prefer to use in each case. The scenarios were:

(1) An artistic VR experience where vibrotactile effects would be created across the body.
(2) An exercise video game where players receive vibrotactile feedback.
(3) A demonstration of force-feedback haptics to the general public using a 1 degree-of-freedom (DoF) device.

These scenarios were designed to focus on common haptics applications and to have multiple plausible answers. Following this task, participants were asked how they selected each tool and how they would improve the taxonomy, such as by adding or revising labels in it.

Twelve of these participants[2] continued to the second activity: an hour-long semi-structured interview. Different versions of the interview were conducted depending on the background of the participant. The six participants with experience *creating* haptic authoring tools were asked to discuss the characteristics of authoring tools they find most impactful, both in their work and to the broader haptics community (creator version). The other six participants were asked to describe challenges they encountered

[1]The questionnaire and interview guides for the activities are included as supplemental materials.

[2]Participant 1 could not be invited to the second activity due to a lack of contact information.

Table 2: Definitions for values of *Effect Localization, Design Approaches*, and *UI Metaphors*.

Attribute	Value	Definition
Effect Localization	Device-centric	Control is leveraged directly over the actuators. The tool has no information about where an effect will be felt.
	Location-aware	The desired location of an effect is taken into account, but presented through the context of the device.
	Target-centric	Effects are directly authored on a specific part of the body or a virtual object. How the stimuli are created on the device is abstracted away.
Design Approaches	Direct	The user directly controls low-level features of an effect, such as the frequency or intensity of vibration, or the location and intensity of force-feedback.
	Procedural	Mappings and algorithms, such as physical models, interpolation, and audio-to-haptic conversion, can be employed to control low-level features.
	Additive	Pre-existing effects can be combined to create larger, more complex experiences.
	Library	Example effects may be selected and used as they are or after additional modification.
	Description	Effects are retrieved or produced using natural language descriptions of how they would be perceived.
UI Metaphors	Track	A timeline represents an interactive channel containing effects, similar to those in audio or video editing.
	Keyframe	Key point for the effect are set and behaviour between them is interpolated.
	Music Notation	An adaption of a musical score or other notation is used to represent haptic effects.
	Dataflow	A dataflow programming model controls haptic output.
	Demonstration	A simple mapping connects physical actions (e.g., taps on a button) or other input data to the features of the haptic effect (e.g., the timing of vibration).

Table 3: Information on the study participants

ID	Experience (years)	Background	Interview Version
1	2	Self-described novice	N/A
2	2	Tactile perception	Searcher
3	2	Human-robot interaction, interest in haptics	Searcher
4	9	VR/simulation haptics	Creator
5	4	Texture rendering	Creator
6	40	Accessibility and haptics	Creator
7	9	Ultrasound haptics	Creator
8	2	Vibrotactile haptics	Searcher
10	17	Music technology and industrial haptics	Creator
11	15	Music technology and VR haptics	Creator
13	1	Music technology and ultrasound haptics	Searcher
14	1.5	Texture rendering	Searcher
16	3	VR haptics	Searcher

applying the taxonomy in the first activity and their past experience *searching* for authoring tools in their own projects (searcher version).

Participants were compensated CAD 10 and CAD 20 upon completion of the first and second activities respectively. Their responses across the two activities were reviewed, coded, and synthesized by the first and second authors to identify characteristics relevant when creating or selecting haptic authoring tools, including:

- the ability for a tool to connect to new software or hardware;
- whether a tool emphasizes the signal produced at an actuator or what is perceived by a user;
- whether stimuli can be synchronized across the senses;
- related work between tools;
- which hardware is specifically supported by the tool; and

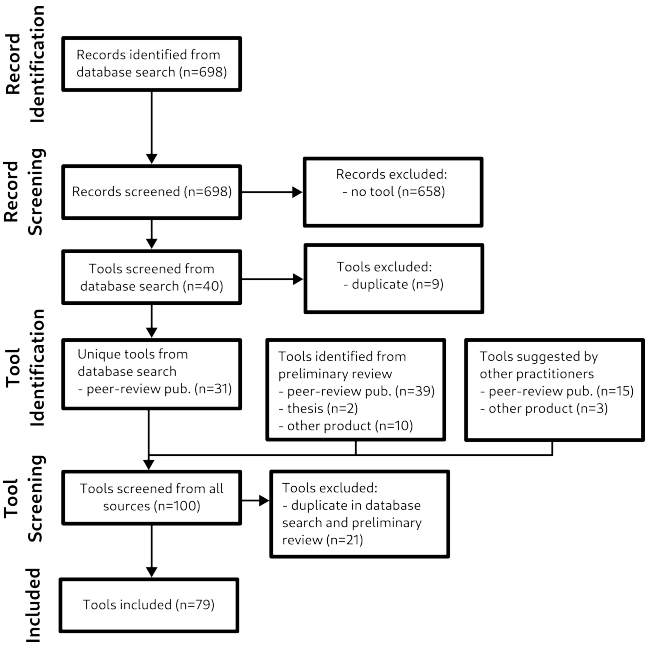

Figure 1: The process through which we identified, screened, and included different records describing haptic authoring tools. These are presented using an adapted version of the PRISMA flow diagram [73].

- to which part or parts of the body haptic effects are delivered.

5 Literature Review

Following the evaluation of the initial taxonomy, a systematic literature review was conducted (Figure 1). In late 2024, we searched the Inspec database for the terms "haptic authoring tool" and "haptic prototyping tool." Searches attempted with other terms, such as "haptic design tool," yielded too large a number of irrelevant results to be included. In the specific example given, "design" yielded numerous generic results, while "tool" frequently matched publications describing hardware devices. Using more restrictive terms, such as "vibrotactile authoring tool," returned a small number of items of which we were already aware.

The database search yielded 698 records, of which, after screening of the abstract and full text, 658 were found to *not* describe a haptic authoring tool. The remaining 40 records corresponded to 31 unique haptic authoring tools, 21 of which were already present in the initial collection. As discussed further in Section 8.4, there are several reasons why these tools may not have been discovered through the literature review. A further 18 tools were suggested directly to the authors, most often by hapticians participating in the study to evaluate the taxonomy. The 10 tools from the database search and 18 tools from suggestions were added to the 51-tool initial collection, resulting in the current collection of 79 tools.

The tools in the updated collection were (re-)coded by the first author following the method described in Section 3.

6 Current Taxonomy

Refinements were made to the initial taxonomy to increase its evaluative and descriptive power. Seven new attributes were added to capture characteristics deemed important by participants in the study:

(1) *Venues* communicates the intended audience of a tool.
(2) *Use Case* communicates the intended purpose of a tool.
(3) *Device Names* supplements *Category* by specifying the hardware a tool supports.
(4) *Device Template* supplements *Abstraction* by indicating if new hardware can be supported through template or configuration files.
(5) *Body Positions* lists where on the body the tool explicitly supports delivering haptic effects. This support could take the form of an egocentric authoring interface or compatibility with wearable hardware such as vests, bracelets, and shoes.
(6) *Storage* lists how effects created with the tool can be saved and shared.
(7) *Connectivity* details methods by which the tool can be connected to new software at runtime.

The permitted values for *Category* and *Non-Haptic Media* were also updated to better describe uncommon features of tools not in the initial collection, such as the support for electrotactile hardware in TactTongue [64] or for olfactory stimuli in the tool made by Coelho et al. [15]. These new and revised attributes of the taxonomy are included in the overview in Table 4. Appendix A describes two example tools using the taxonomy in order to help the reader understand how it can be applied.

This version of the taxonomy was applied to the 79 haptic authoring tools in the current collection, the results of which are included as supplemental materials and serve as the basis for the descriptive analysis in Section 7.

7 Descriptive Analysis

In this section, we use the taxonomy to structure an analysis of trends in the development of haptic authoring tools present in our collection. This analysis is intended to demonstrate the taxonomy's *descriptive* capacity [4] and to familiarize practitioners with the haptic authoring tool literature. The five most common values for each Metadata, Hardware, and Interaction attribute are shown in Appendix B in Tables 6, 7, and 8 respectively.

7.1 Metadata

While our collection includes tools released between 1995 [10] and 2024 [43], the majority were only published during or after 2010. With fewer than ten tools found that preceded 2007, it seems that interest in the topic of haptic authoring has increased over time. The goals of these projects were quite varied; more than a dozen use cases were identified when applying the taxonomy. The most common purposes of a tool were:

- to serve as a prototype for future authoring tools (21, e.g., [38, 84, 105]);
- to control a particular hardware device (15, e.g., [27, 40, 62]);
- to augment non-haptic media with haptic content (15, e.g., [22, 26, 47, 70]);

Table 4: Overview of the current taxonomy.

	Attribute	Definition	Values	Values per Tool
Metadata	Release Year	The year the tool was released or described in a publication.	Any year	1
	Availability	Whether the tool is available to the public.	Available | Unavailable	1
	License	The terms under which the tool is/was available, if known.	Open Source | Proprietary | Non-Commercial | Unknown	1
	Use Case	The purpose for which the tool was intended.	Any purpose	1 or more
	Platform	The software platform(s) that run the tool.	Any name	1 or more
	Venues	Academic venues where the tool was described.	Any venue	0 or more
Hardware	Category	The type of hardware device(s).	Vibrotactile | Force-feedback | Skin stretch | Temperature | Electrotactile	1 or more
	Device Names	Explicitly supported device(s).	Any device	1 or more
	Abstraction	Whether support extends to more specific or generic categories of device.	Custom | Consumer | Class	1
	Device Template	Whether support can be easily extended to new types of devices.	Yes | No	1
	Body Positions	Where on the body hardware for the tool is coupled, if applicable.	Arm | Foot | Hand | Head | Leg | Tongue | Torso	0 or more
Interaction	Driving Feature	Whether effects change due to time or miscellaneous actions.	Time | Action	1 or more
	Effect Localization	The extent to which effects made with the tool are mapped to an actuator or a separate target.	Device-centric | Location-aware | Target-centric	1
	Iterative Playback	Whether effects can be rendered within the tool, if applicable.	Yes | No | N/A	1
	Non-Haptic Media	Whether non-haptic media is supported in any way.	Audio | Visual | Olfactory	0 or more
	Design Approaches	The method(s) available in the tool to create an effect.	Direct | Procedural | Library | Additive | Description	1 or more
	UI Metaphors	Common patterns which define user interaction with the tool.	Track | Keyframe | Music Notation | Dataflow | Demonstration	0 or more
	Storage	Method(s) by which effects are saved/loaded.	Any method	0 or more
	Connectivity	Methods to connect new data or software.	Any method	0 or more

- to design haptic interactions for virtual reality (13, e.g., [23, 24, 71, 93]);
- to create haptics for entertainment, such as music or video games (13, e.g., [34, 37, 60]); and
- to simulate real physical experiences (11, e.g., [16, 19, 77]).

Other use cases, such as accessibility [64, 66, 75], education [63], and psychophysical experimentation [28, 65] were also described.

Tools vary considerably in the software platform they require to run. The plurality of tools are capable of running on common desktop operating systems, such as Windows, macOS, or Linux. Since the increase in research on authoring tools in the 2010s, web-based [6, 25, 85, 89] and smartphone-based [29, 38, 41, 84] projects have become quite common. The Unity game engine[3] is a frequent choice for tools intended for use in virtual reality [23, 24, 43, 95].

Publications about authoring tools were found in 40 journals and conferences, broadly grouped into the topics of haptics (e.g., Haptics Symposium [19, 28, 70, 76, 84, 85, 96, 97], World Haptics [12, 42, 52, 83, 89], EuroHaptics [5, 16, 26]), human-computer interaction (e.g., CHI [40, 41, 43, 45, 46, 86], UIST [22, 24, 64, 93, 109], TEI [38, 101, 106]), and sound and music (e.g., NIME [31, 49, 90], ICMC [10, 11, 37], HAID [30, 53]). Tools developed for less common use cases, such as to study human perception [65] or explore haptic multimedia [1, 47], were published in venues related to those topics. Two tools,

[3] https://unity.com/

Skinscape [34] and the Tactile Editor [44], appear to have only been discussed in the context of theses.

Hapticians interested in finding a tool to use in their own work face a significant problem: 40 of the 79 tools in our collection were not available to the public (Figure 2). The majority of these tools were seemingly never intended for public use, and were instead built for a specific project [10, 22, 23, 37] or research objective [38, 46, 83, 99]. Of the 39 available tools, 27 are open-source [16, 43, 62, 65, 100], five are proprietary [3, 6, 20], one is non-commercial [19]. The remaining six available tools have source code published, but do not include a license granting rights to users, leaving their status ambiguous [59, 63, 85].

It is unclear how the problem of tool unavailability will evolve in the coming years. Since most unavailable tools originate in research projects, if and how future researchers release their tools is likely to determine the extent to which haptics practitioners can access new innovations in haptic authoring. If a greater number of researchers release their tools under open-source terms, as has been the trend since 2020 [24, 27, 30, 31, 43, 49, 62, 65, 78, 93, 100, 102, 105], it will likely become easier for practitioners to find tools suitable for their own work. The haptics community could increase the likelihood of such an outcome by encouraging researchers to produce open-source software that is designed and documented sufficiently for it to be used after the end of its original project.

7.2 Hardware

7.2.1 Types of haptic feedback. Authoring tools generally focus on vibrotactile (54 tools, e.g., [29, 46]) and force-feedback (26 tools, e.g., [28, 102]) effects; temperature [23, 48], skin-stretch [23, 48], electrotactile [64], and other types of haptic feedback receive far less attention. Typically, authoring tools focus on a single type of effect, with only six supporting some combination of vibrotactile, force-feedback, and temperature haptics [12, 13, 22, 23, 39, 48].

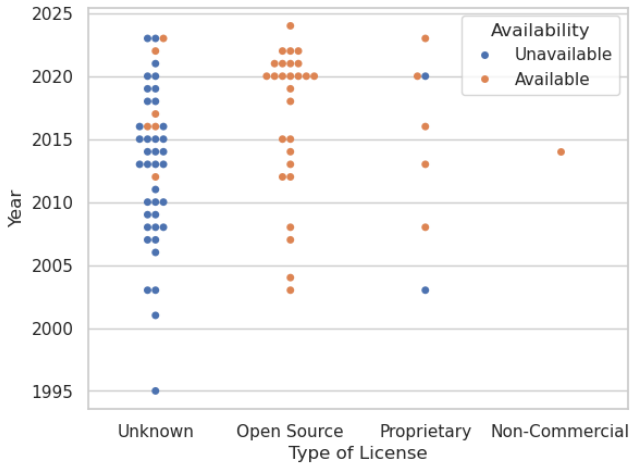

Figure 2: Open-source projects are the most common type of tool that are available to the public, and have become more common since 2020.

7.2.2 Haptic devices. Vibrotactile authoring tools were the most frequent to control classes of hardware [28, 85, 100], or custom devices [46, 47, 93, 105] (Figure 3). Force-feedback tools typically were built for hardware available to consumers, particularly the Phantom and Falcon [5, 19, 26, 91]. Only five tools—four vibrotactile [48, 52, 65, 74] and one force-feedback [16]—provide templates that users can modify to add support for new kinds of devices.

7.2.3 Body positions. Relatively few tools provide explicit functionality for creating haptic effects on specific parts of the body (Table 7). When this support is present, it is usually limited to a single location (Figure 4), such as the hands [12, 24, 43, 47, 50, 65, 71, 86, 95, 99], arms [29], head [46], torso [69], foot [93], or tongue [64]. Several tools include some degree of support for rendering effects on multiple parts of the body, though how this functionality manifests varies. For example, Compressables supports separate devices for different body parts [27], HFX Studio uses a perceptual model of

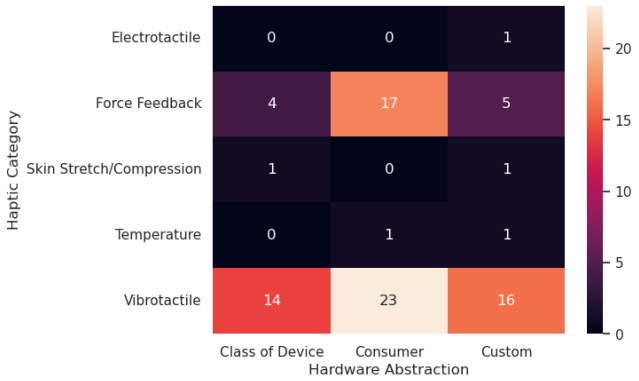

Figure 3: Consumer hardware was the most common type supported by tools, although vibrotactile tools often support classes of devices and custom hardware.

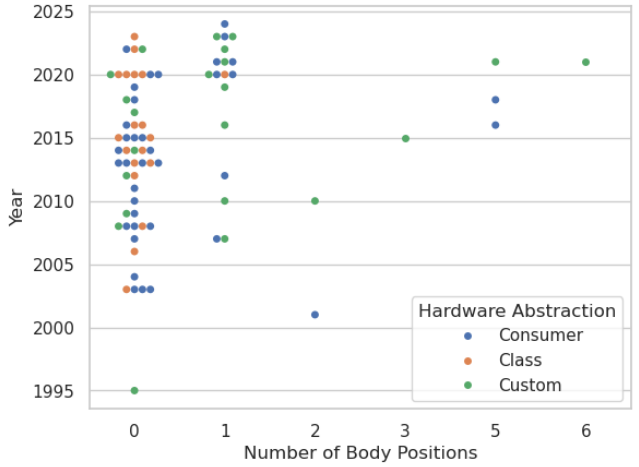

Figure 4: Few tools provide explicit support for authoring effects on different parts of the body. Often the hardware used for this purpose is custom-built.

the human body to determine how to render effects throughout if suitable hardware is connected [23], and TactJam simply provides an interface to position actuators anywhere on a 3D model of the body [105].

7.3 User interface

7.3.1 Spatialization of haptic effects. The user interfaces of the tools had different characteristics, but certain approaches were common. For example, most vibrotactile authoring tools, such as Macaron [85] and the TouchCon Editor [48], permitted users to design effects to be rendered on discrete actuators. These "device-centric" tools do not include functionality to aid in the creation of spatialized effects (Figure 5). This is less true for other types of haptic effects, particularly force force feedback. Target-centric tools focus on authoring on part of the body [23, 37, 46, 95] or an interactive virtual object [13, 23, 102]. In between these two extremes, various methods are used to take the location of haptic stimuli into account. For tools to control a vibrotactile array, visualizing the location of actuators [47, 60, 69] or including a function to create phantom vibrations [40, 78, 83] are categorized as location-aware. TactJam, with its explicit mapping of particular actuators to body locations that does not in any way impact rendering, is also an example of a location-aware tool [106].

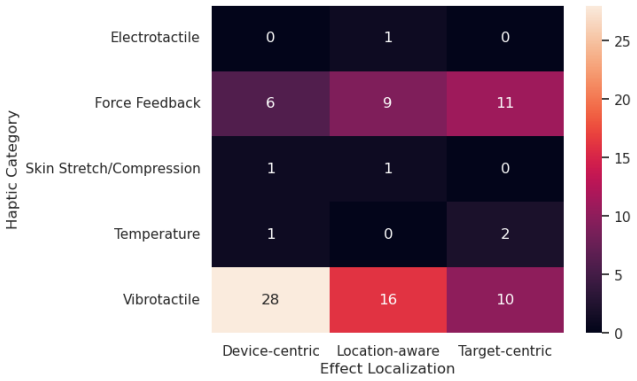

Figure 5: Vibrotactile authoring tools often make the actuator, rather than an object or part of the body, the focus of the design.

7.3.2 Creating haptic stimuli. While the five *Design Approaches* (Table 8) describe how tools can be used to create haptic effects at a high level, how they are implemented varies. For example, Direct editing functionality permits users to control the frequency and intensity of tactile stimuli [66, 84, 85] or the location and intensity of force feedback [12, 28, 100]. Procedural methods of manipulating such low-level features include interpolation [66, 68, 96], physical models [5, 11, 77], and audio-to-haptic translation [56, 59]. Effects can be combined (Additive) as a sequence in time [23, 52, 96, 97] and across multiple devices [34, 66, 80, 106]. Examples provided in a tool's Library can be used as they are or after additional modification [41, 62, 97], and in certain cases they can be produced by other users [106]. Descriptions are used to access pre-existing effects [64, 89] or produce new ones [36, 82]. Authoring tools have

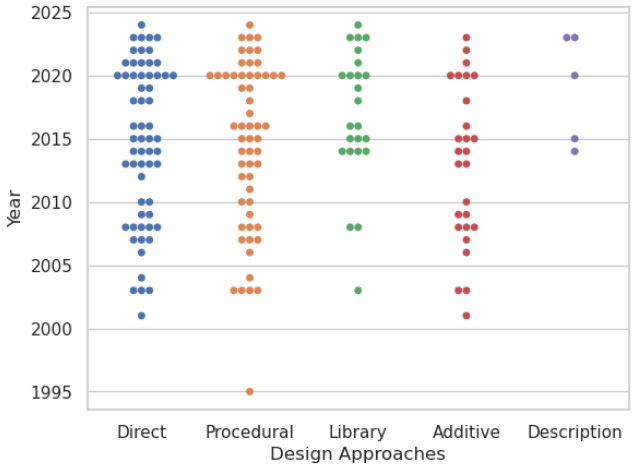

Figure 6: The design approaches present in tools over time.

included Library and Description functionality more often in the last decade, while the other approaches have been consistently present (Figure 6).

It is typical for tools to have functionality corresponding to multiple *Design Approaches*; 26 tools in the collection include two approaches [46, 49, 89], 21 tools include three [43, 93, 109], and nine tools include four [23, 80, 100]. Generally, the more popular approaches are implemented together in authoring tools. Direct and Procedural were the most common pair [24, 95] with combinations including the Additive [22, 69] and Library [62, 106] approaches following.

In nearly every tool that includes an interactive interface (e.g., [59, 66, 74, 83]), it is possible for users to iteratively test and improve haptic effects. Most of the remaining tools, such as the Penn Haptic Texture Toolkit [19] and CHAI3D [16], are software libraries or otherwise used in a manner where iterative playback is not applicable. Only three tools [5, 38, 71] have interactive interfaces and appear to lack functionality to test a design without leaving the editing context.

When deployed, an effect can be triggered or change as a function of time, in-application actions, or both (Figure 7). Actions include discrete events, such as in-game events in FeelCraft [82], and continuous properties, such as the position of the end effector of a force-feedback device [19], tracking a hand to produce vibrations upon it [43, 65, 86], or the properties of vocalizations the designer makes [59]. Exclusively time-based design is only present in vibrotactile tools and the one electrotactile tool. Exclusively action-based design is predominantly present in force-feedback tools, with FeelCraft [82], PhysVib [77], and vibrAteRial [93] being exceptional action-based vibrotactile tools.

Despite haptic content often being destined for use in a multisensory context [58], 51 of the 79 tools had no support for displaying, creating, or editing non-haptic media. Of those remaining, all but one focused on audio [2, 5, 10, 17, 31, 34, 56, 67], visual [12, 26, 32, 35, 48, 70, 71], or audio and visual media [1, 13, 16, 22, 23, 47, 53, 86, 91, 97, 108]. The remaining tool, developed by Coelho et al., supports viewing 360° video and incorporating

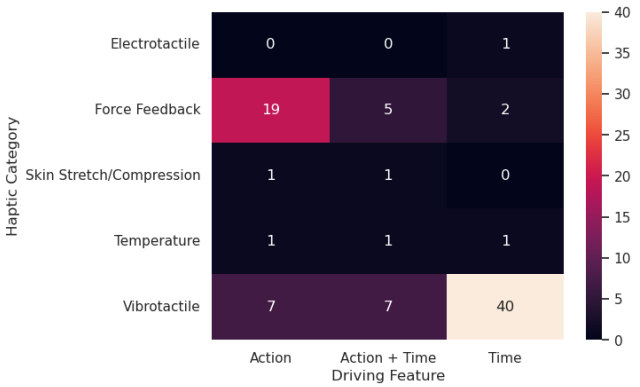

Figure 7: A summary of whether tools for different types of haptic effects support time-based and action-based design.

additional audio, haptic, and olfactory stimuli [15]. Generally, non-haptic media is only included for synchronization purposes, but certain music-focused physical modeling tools produce sound and force feedback simultaneously [5, 53].

7.3.3 Common UI metaphors. In tools with a graphical user interface (62 of 79), patterns in how design approaches were made available to the user were tracked under *Interaction Metaphors*. While 14 tools, such as FeelCraft [82] and HAMLAT [26], exclusively used standard menu items such as sliders and checkboxes, the remaining 48 tools incorporated interaction elements frequently used in other domains. Drawing from audio and visual editing software, 24 tools added the concept of a track to present an interactive timeline of haptic effects. Different implementations of this feature could be used to sequence effects [34, 43, 66, 83, 96], synchronize multisensory stimuli [1, 97], and direct stimuli to different devices [48, 66, 83]. A related feature from visual editing present in 22 tools was keyframes, which allow for key points for an effect to be set in time [12, 42, 46, 83, 85] or space [96], and for in-between values to be interpolated. Two tools, VibScoreEditor [42] and Beadbox [66], adapted musical notation to author complex vibrotactile effects.

In 27 tools, functionality was present to allow users to "demonstrate" how an effect should feel or where it should occur. This could take various forms, such as tapping or sketching vibrotactile patterns via an input device [3, 38], recording the position of a force-feedback device [12, 96], and tracing where an effect should be felt on the body [37, 86, 95]. This feature could also take the form of capturing and transforming input in another modality, such as from real-time vocalizations [59] or recorded audio [56, 63]. Finally, forms of logic could be implemented via a dataflow programming interface in six tools [2, 5, 11, 30, 75, 100].

7.3.4 Interoperability. Little consensus was observed in terms of functionality to store haptic experiences or to connect to other software at runtime. Of the 55 tools with some functionality for saving, importing, and exporting data, 24 used custom XML, JSON, or CSV files as a data storage format [43, 65, 85, 93]. Audio files were used in several vibrotactile authoring applications [56, 63, 78, 85]. The MPEG container format was used to save haptic data in three cases [12, 25, 47]. Generally, files written with one tool were unlikely to work in another, with the exception of single-channel vibrotactile patterns exported as audio.

The majority of tools, 45 of 79, have no functionality for connecting to other software. An API to access core authoring functionality [16, 35, 78] or to play back effects created with the tool in another environment [25, 60, 100] was included in 22 tools. Network protocols, primarily based on Open Sound Control, were used in eight tools for communication with or between parts of the tool [31, 37, 40, 43, 44, 49, 91]. Such a protocol could be leveraged to add functionality to a system in ways not foreseen by its creators.

8 Discussion

Using the previous descriptive analysis, we identify different general gaps in the current haptic authoring literature. In real-world use, we anticipate that practitioners would delve deeper than we do here, using the taxonomy, descriptive analysis, and documents in the authoring tool collection to generate novel research directions and effective solutions to the challenges they face. Following this identification of gaps, we discuss the extent to which requirements for authoring tools identified by previous authors are met by the tools in our collection, and the suitability of our taxonomy for use by haptics practitioners.

8.1 Problems and opportunities in haptic authoring tools

8.1.1 Tools for different haptic devices. The vast majority of authoring tools we found are intended for vibrotactile (54 tools) and force-feedback devices (26 tools). Within the vibrotactile category, tools typically focus on controlling actuators as part of wearables [6, 37, 93], embedded arrays [69, 78, 83], or in isolation [84, 85, 89]. Commercial hardware for VR environments [23, 24] or ultrasound haptics [43, 65, 86] are also well represented in the collection. For force-feedback, desktop-sized one-DoF [5, 28, 96, 100] or commercial higher-DoF devices [12, 16, 76, 91] have received the most attention.

Researchers could work to improve haptic authoring with non-commercial and more complex devices. Different approaches to haptic authoring could be developed for other forms of haptic interaction, including temperature, skin stretch, or electro-muscle stimulation, to help make these technologies more accessible to haptics and HCI practitioners. Opportunities also exist within vibrotactile and force-feedback authoring. For example, vibrotactile authoring tools could include functionality to aid in embedding actuators into physical objects [9], and force-feedback tools could focus on interaction on different physical scales. Cobity, for example, is designed to create room-scale haptic interactions for virtual reality environments rather than the typical desktop scale [102].

8.1.2 Scarcity of multimodal authoring tools. Support for non-haptic media is present in few haptic authoring tools and is generally limited to synchronization (Section 7.3.2). This way of presenting media for other sensory modalities imposes a specific model of haptic design on the user of the tool. For the non-haptic media to be useful for synchronization, it must be imported in a nearly final state, meaning that haptic design must occur *after* other forms of design. Because it is generally impossible to edit non-haptic media

within the tool, users are implicitly encouraged to iteratively refine haptic aspects of the interaction and keep the rest unchanged.

While this approach to multisensory design may be successful for some, it is not the only viable option. For example, research on audiovisual practitioners found that people engaged in one of two approaches: dialogic, where one sensory modality leads the other, and synchronous, where they are designed simultaneously [104]. Supporting a synchronous approach to multimodal haptic design may be an interesting direction for future research in haptic authoring tools, possibly resulting in software that is better suited to different tasks or practitioner preferences than what currently exists.

8.1.3 Alternative methods of user interaction. The same few design approaches (direct, procedural, additive) and UI metaphors (track, keyframe, demonstration) appear in haptic authoring tools (Table 8). While these methods have clearly been successful, additional research into alternatives, including those identified in our collection, may be worthwhile. For example, dataflow- and music notation-based user interfaces are scarcely studied in haptics, despite dataflow interfaces being present in popular audio programming tools such as Max MSP [4] and Pure Data [79], and music notation being used for decades to describe vibrotactile effects [8].

Haptic authoring tools using natural language description have become more popular in recent years, and will likely continue to be developed as large language models make them easier to implement. Since our literature review was conducted, descriptions of vibrotactile effects have been at the core of tools such as ChatHAP [55] and HapticGen [94].

8.1.4 Interoperability and reusable features between tools. Interoperability between authoring tools and other software was a key characteristic that was identified through our study with haptics practitioners. Despite this, users of haptic authoring tools are largely locked into a specific environment. Haptic effects and interactions designed with one tool are generally not usable with other ones, making switching from one tool to another difficult. Adding new functionality or supporting new interactions is also challenging; only 22 tools include an API that could allow it to be used in new ways (Section 7.3.4).

The recent MPEG haptics coding standard [33] may be used in future tools to allow for haptic effects to be reused more easily. If not, additional efforts to develop an interchange format for haptics will certainly be warranted.

Research to build haptic authoring tools with expandability in mind may be another worthwhile direction. Based solely on support for different types of hardware (Section 8.1.1), there are myriad projects involving haptics where no suitable tool exists. If the functionality in tools were built in a modular way, possibly inspired by frameworks for audio plug-ins such as LV2,[5] it could reduce the burden in creating authoring tools, or developing alternative interfaces for them as we mention in Section 8.1.3.

[4]https://cycling74.com/products/max
[5]https://lv2plug.in/

8.2 Adherence to authoring tool guidelines

As mentioned in Section 2.2, the requirements formulated by Schneider et al. [83], Seifi et al. [86], and van Oosterhout et al. [100] could be interpreted as broad guidelines for haptic authoring tools. Real-time playback, the saving and loading of content, and libraries of presets were included in all three sets of requirements. In addition, the ability to configure new types of hardware to work with the tool [83, 100] and to synchronize with non-haptic media [83, 86] were each included in two of the sets and described as important features of authoring tools by expert hapticians in the interviews in Section 6. These five guidelines map well to fields of the taxonomy: real-time playback to *Iterative Playback*, saving and loading content to *Storage*, synchronization with non-haptic media to *Non-haptic media*, presets to the library label of *Design Approaches*, and configuration of new hardware to *Device Template* and *Hardware Abstraction*. Using the information on the tools in the collection, we can roughly assess how well existing authoring tools meet these guidelines.

Interestingly, the guidelines are followed at vastly different rates. For example, real-time playback is consistently present in authoring tools as reflected by *Iterative Playback*. Similarly, values for *Storage* indicate that nearly three quarters of the tools currently captured in the collection have some mechanism for importing or exporting data. In contrast, only a minority of tools include any support for non-haptic media. When considering only those tools that are presently available, only eight provide such support. While libraries of presets are more common, with 24 of 79 tools and 16 of 39 available tools including the library *Design Approach*, these still represent a minority of tools in our dataset.

The situation is more complex when assessing the degree of support for new hardware. Only five tools have template code for supporting completely different kinds of devices listed as part of *Device Template*, but far more tools include support for classes of devices under *Hardware Abstraction*. Tools to control audio-driven vibrotactile devices are particularly flexible in terms of the hardware they can use and how it is incorporated into computer interactions. Due to the variety of methods by which a tool could be made to support new hardware, a full assessment of adherence to this guideline would require an investigation outside the scope of this work. However, it appears to occur in a similar number of tools in our dataset as the guideline for presets.

It may be of interest to haptics practitioners to investigate why implementation of these features varies. Differences in adoption suggest that certain guidelines may be more important or easier to implement than others, justifying the need for revised guidelines or additional resources to help the creators of authoring tools.

8.3 Suitability of the taxonomy

To assess if our taxonomy is likely to be suitable for use by practitioners, we verify here the extent to which we have demonstrated its descriptive, evaluative, and generative power [4].

8.3.1 Descriptive power. Our taxonomy is capable of representing meaningful differences between 79 tools for different hardware and applications developed over a thirty year period. As our tool collection expanded as a result of suggestions and our literature

review, the taxonomy continues to be capable of describing key differences between the tools.

8.3.2 Evaluative power. The taxonomy was successfully used by participants in the study in Section 4 to compare between authoring tools and select one for a specific purpose. While it would be useful to study how our taxonomy is used by haptics and HCI practitioners with real problems to further improve it, these practitioners would need to have the taxonomy and a set of tools labeled with it. Since this is not possible until the taxonomy is made published, we consider the study to be a sufficient demonstration of its evaluative power for this time.

8.3.3 Generative power. In Section 8.1, the taxonomy is used to identify several gaps in haptic authoring tool research. Due to the broad focus of our work, there is certainly nuance within certain families of tools, such as single-actuator vibrotactile authoring tools, that is not captured in the taxonomy. However, we believe that this scope provides useful insights into the field and helps identify opportunities that would otherwise be missed. As such, we believe that the taxonomy has sufficient generative power, and meets all the evaluative criteria.

8.4 Future work

Regardless of the utility of our taxonomy and descriptive analysis, there remains room for additional work on this subject. Our evaluation included only practitioners with prior experience using haptic authoring tools due to our focus on aiding this user community. Additional studies could be conducted with those interested in haptics who have never used an authoring tool to identify and better meet their needs.

Since there are not standard keywords to describe what we call "haptic authoring tools," alternative database searches may have identified projects not included in our collection. In general, tools can be missed by searches because they are often developed and published in the context of a larger research project (e.g., vibrAteRial in bARefoot [93]). This also does not touch on the challenge of locating the many commercial tools without a corresponding publication. As a result, our collection is not comprehensive and our taxonomy may not adequately describe every existing tool. Further, as innovations continue to be made in haptic authoring, new information that is not adequately captured by our taxonomy will undoubtedly become important for practitioners to know.

In response to this situation, we hope that the taxonomy, analysis, and collection contributed in this paper will provide a base upon which future work can be conducted. Our website is open to contributions[6] in the form of new tools, updated information, and suggested revisions to the taxonomy in order to simplify this. Additionally, we encourage others to construct application-specific taxonomies of haptic authoring tools to capture any nuances missed in our general taxonomy.

[6]A guide with instructions on how to contribute is available at https://github.com/JRegimbal/hat-box/blob/main/CONTRIBUTING.md.

9 Conclusion

Authoring tools facilitate haptic design for experts and novices alike. Despite the progress made through decades of research, haptic authoring is not free from problems. Tools do not exist for many devices or applications, nor are they suited to the needs of every kind of user. Even in cases where considerable work has already been done, such as vibrotactile authoring, haptics and HCI practitioners are often not aware of relevant work. Consequently, they are unable to benefit from what has been done by either using existing authoring tools as they are or by using past research to inform future projects.

To address this problem, we conducted a systematic literature review of haptic authoring tools and a study with practitioners to develop a haptic authoring tool taxonomy. Using this taxonomy, we were able to perform a descriptive analysis of the tools found through our review and identify several gaps in the current literature that may be of interest to the haptics and HCI communities. We have also created an online version of this taxonomy and collection, which we call the Haptic Authoring Toolbox or HAT Box,[7] to serve as an interactive additional resource to this paper. We anticipate that the results of this work will guide these practitioners in their efforts to advance the state of the art in haptic design.

Acknowledgments

This work was supported by the Natural Sciences Engineering Research Council of Canada under Canadian Graduate Scholarship (Doctoral) no. 569236, by the Fonds de recherche du Québec—Nature et technologies under doctoral scholarship no. 315050, and by a Vadasz McGill Engineering Doctoral Award to the lead author.

The authors thank Jeffrey Blum, Antoine Weill–Duflos, Pascal Fortin, Yongjae Yoo, Hasti Seifi, Sri Gannavarapu, Rayan Isran, and the members of the Shared Reality Lab for their feedback over the course of this project. We also thank the participants in our study for their time and insights.

References

[1] Md. Abdur Rahman, Abdulmajeed Alkhaldi, Jongeun Cha, and Abdulmotaleb El Saddik. 2010. Adding haptic feature to YouTube. In *Proc. ACM Int. Conf. Multimedia*. 1643–1646. doi:10.1145/1873951.1874310

[2] Actronika SAS. 2020. Actronika Evaluation Kit v0.1.6. Retrieved 2025-11-12 from https://www.youtube.com/watch?v=Od3ZZCSfHwQ

[3] Apple. 2018. Change iPhone sounds and vibrations. Retrieved 2024-09-27 from https://support.apple.com/en-ca/guide/iphone/iph07c867f28/12.0/ios/12.0

[4] Michel Beaudouin-Lafon. 2004. Designing interaction, not interfaces. In *Proceedings of the working conference on Advanced visual interfaces (AVI '04)*. Association for Computing Machinery, New York, NY, USA, 15–22. doi:10.1145/989863.989865

[5] Edgar Berdahl, Peter Vasil, and Andrew Pfalz. 2016. Automatic Visualization and Graphical Editing of Virtual Modeling Networks for the Open-Source Synth-A-Modeler Compiler. In *Proc. Eurohaptics*. 490–500. doi:10.1007/978-3-319-42324-1_48

[6] bHaptics. 2024. bHaptics Designer. Retrieved 2026-01-18 from https://designer.bhaptics.com/

[7] Arpit Bhatia, Moaaz Hudhud Mughrabi, Diar Abdlkarim, Massimiliano Di Luca, Mar Gonzalez-Franco, Karan Ahuja, and Hasti Seifi. 2025. Text Entry for XR Trove (TEXT): Collecting and Analyzing Techniques for Text Input in XR. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–18. doi:10.1145/3706598.3713382

[8] Stephen Brewster and Lorna M. Brown. 2004. Tactons: structured tactile messages for non-visual information display. In *Proceedings of the fifth conference on*

[7]https://srl.mcgill.ca/hat-box

Australasian user interface - Volume 28 (AUIC '04). Australian Computer Society, Inc., AUS, 15–23.

[9] Aditya Tirumala Bukkapatnam, Philippe Depalle, and Marcelo M. Wanderley. 2020. Defining a vibrotactile toolkit for digital musical instruments: characterizing voice coil actuators, effects of loading, and equalization of the frequency response. *J Multimodal User Interfaces* 14, 3 (Sept. 2020), 285–301. doi:10.1007/s12193-020-00340-0

[10] Claude Cadoz, Jean-Loup Florens, and Annie Luciani. 1995. Musical Sounds, Animated Images with CORDIS-ANIMA and its Multimodal Interfaces. In *Int. Comput. Music Conf.* 531–532.

[11] Nicolas Castagné and Claude Cadoz. 2002. GENESIS : a Friendly Musician-Oriented Environment for Mass-Interaction Physical Modeling. In *Int. Comput. Music Conf.*, Ann Arbor (Ed.). 330–337.

[12] Jongeun Cha, Yongwon Seo, Yeongmi Kim, and Jeha Ryu. 2007. An Authoring/Editing Framework for Haptic Broadcasting: Passive Haptic Interactions using MPEG-4 BIFS. In *World Haptics Conf.* 274–279. doi:10.1109/WHC.2007.20

[13] Liwei Chan, Mu-Hsuan Chen, Wen-Chi Ho, Roshan Lalintha Peiris, and Bing-Yu Chen. 2019. SeeingHaptics: Visualizations for Communicating Haptic Designs. In *Proc. Int. Conf. Human-Comput. Interact. Mobile Devices Services*. 1–5. doi:10.1145/3338286.3340112

[14] S. Choi and K. J. Kuchenbecker. 2013. Vibrotactile Display: Perception, Technology, and Applications. *Proc. IEEE* 101, 9 (Sept. 2013), 2093–2104. doi:10.1109/JPROC.2012.2221071

[15] Hugo Coelho, Miguel Melo, Luís Barbosa, José Martins, Mário Sérgio, and Maximino Bessa. 2018. Immersive Edition of Multisensory 360 Videos. In *Proc. World Conf. Inf. Syst. Technol.* 309–318. doi:10.1007/978-3-319-77712-2_30

[16] F. Conti, F. Barbagli, R. Balaniuk, M. Halg, C. Lu, D. Morris, L. Sentis, E. Vileshin, J. Warren, O. Khatib, and K. Salisbury. 2003. The CHAI Libraries. In *Proc. Eurohaptics*. 496–500.

[17] Immersion Corporation. 2015. Haptic Studio.

[18] D Cuartielles, A Göransson, T Olsson, and S Stenslie. 2012. Developing Visual Editors for High-Resolution Haptic Patterns. In *7th Int. Workshop Haptic Audio Interact. Des.* Lund, Sweden, 42–45.

[19] H. Culbertson, J. J. López Delgado, and K. J. Kuchenbecker. 2014. One hundred data-driven haptic texture models and open-source methods for rendering on 3D objects. In *2014 IEEE Haptics Symp.* 319–325. doi:10.1109/HAPTICS.2014.6775475

[20] D-BOX. 2024. Live Motion SDK License. Retrieved 2024-09-27 from https://www.d-box.com/en/license-live-motion-sdk

[21] Tor-Salve Dalsgaard and Oliver Schneider. 2025. A Unified Model for Haptic Experience. *ACM Trans. Comput.-Hum. Interact.* 32, 1 (April 2025), 9:1–9:38. doi:10.1145/3711842

[22] Fabien Danieau, Jérémie Bernon, Julien Fleureau, Philippe Guillotel, Nicolas Mollet, Marc Christie, and Anatole Lécuyer. 2013. H-Studio: an authoring tool for adding haptic and motion effects to audiovisual content. In *Proc. Adjunct ACM Symp. User Interact. Softw. Technol.* New York, NY, USA, 83–84. doi:10.1145/2508468.2514721

[23] Fabien Danieau, Philippe Guillotel, Olivier Dumas, Thomas Lopez, Bertrand Leroy, and Nicolas Mollet. 2018. HFX studio: haptic editor for full-body immersive experiences. In *Proc. ACM Symp. Virtual Reality Softw. Technol.* 1–9. doi:10.1145/3281505.3281518

[24] Donald Degraen, Bruno Fruchard, Frederik Smolders, Emmanouil Potetsianakis, Seref Güngör, Antonio Krüger, and Jürgen Steimle. 2021. Weirding Haptics: In-Situ Prototyping of Vibrotactile Feedback in Virtual Reality through Vocalization. In *ACM Symp. User Interface Softw. Technol.* 936–953. doi:10.1145/3472749.3474797

[25] H. Dong, Y. Gao, H. Al Osman, and A. El Saddik. 2015. Development of a Web-Based Haptic Authoring Tool for Multimedia Applications. In *2015 IEEE International Symposium on Multimedia (ISM)*. 13–20. doi:10.1109/ISM.2015.71

[26] Mohamad Eid, Sheldon Andrews, Atif Alamri, and Abdulmotaleb El Saddik. 2008. HAMLAT: A HAML-Based Authoring Tool for Haptic Application Development. In *Proc. Eurohaptics 2008 (Lecture Notes in Computer Science)*. 857–866. doi:10.1007/978-3-540-69057-3_108

[27] Shreyosi Endow, Hedieh Moradi, Anvay Srivastava, Esau G Noya, and Cesar Torres. 2021. Compressables: A Haptic Prototyping Toolkit for Wearable Compression-based Interfaces. In *Designing Interactive Systems Conference*. 1101–1114. doi:10.1145/3461778.3462057

[28] M. J. Enriquez and K. E. MacLean. 2003. The hapticon editor: a tool in support of haptic communication research. In *11th Symp. Haptic Interact. Virtual Environ. Teleoperator Syst.* 356–362. doi:10.1109/HAPTIC.2003.1191310

[29] Juan Pablo Forero Cortes, Hussel Suriyaarachchi, Alaeddin Nassani, Haimo Zhang, and Suranga Nanayakkara. 2021. OM: A Comprehensive Tool to Elicit Subjective Vibrotactile Expressions Associated with Contextualised Meaning in Our Everyday Lives. In *Proc. Int. Conf. Mobile Human-Comput. Interact.* 1–12. doi:10.1145/3447526.3472028

[30] Christian Frisson, Julien Decaudin, Mario Sanz Lopez, and Thomas Pietrzak. 2020. Printgets: an Open-Source Toolbox for Designing Vibrotactile Widgets with Industrial-Grade Printed Actuators and Sensors. In *Haptic Audio Interact. Des.* Retrieved 2023-06-21 from https://lilloa.univ-lille.fr/handle/20.500.12210/56770

[31] Christian Frisson, Mathias Kirkegaard, Thomas Pietrzak, and Marcelo M. Wanderley. 2022. ForceHost: an open-source toolchain for generating firmware embedding the authoring and rendering of audio and force-feedback haptics. In *Int. Conf. New Interfaces Musical Expression*. doi:10.21428/92fbeb44.76cfc96e

[32] Yu Gao, Hussein Al Osman, and Abdulmotaleb El Saddik. 2013. MPEG-V based web haptic authoring tool. In *2013 IEEE International Symposium on Haptic Audio Visual Environments and Games (HAVE)*. 87–91. doi:10.1109/HAVE.2013.6679616

[33] Philippe Guillotel, Yeshwant Muthusamy, Quentin Galvane, Eric Vezzoli, Lars Nockenberg, Iraj Sodagar, Henry Da Costa, Alexandre Hulsken, Gurvan Lecuyer, Matthieu Perreira Da Silva, Franc¸ois Bouffard, Heather Culbertson, Sandeep Kollannur, and David Gueorguiev. 2025. Adding Touch to Immersive Media: An Overview of the MPEG Haptics Coding Standard. *IEEE Transactions on Haptics* (2025), 1–15. doi:10.1109/TOH.2025.3539026

[34] Eric Gunther. 2001. *Skinscape: A Tool for Composition in the Tactile Modality*. Master's thesis. Massachusetts Institute of Technology.

[35] H3D. 2021. H3D API Manual Version 2.4. Retrieved 2024-10-03 from https://h3d.org/wp-content/uploads/2021/02/H3DAPI_2.4/doc/H3D%20API%20Manual.pdf

[36] Waseem Hassan, Arsen Abdulali, and Seokhee Jeon. 2020. Authoring New Haptic Textures Based on Interpolation of Real Textures in Affective Space. *IEEE Trans. Ind. Electron.* 67, 1 (Jan. 2020), 667–676. doi:10.1109/TIE.2019.2914572

[37] Ian Hattwick, Ivan Franco, Marcello Giordano, Deborah Egloff, Marcelo M Wanderley, Valerie Lamontagne, Ian Arawjo, Chris Salter, and Maurizio Martinucci. 2015. Composition Techniques for the Ilinx Vibrotactile Garment. In *Int. Comput. Music Conf.*

[38] Kyungpyo Hong, Jaebong Lee, and Seungmoon Choi. 2013. Demonstration-based vibrotactile pattern authoring. In *Proc. Int. Conf. Tangible, Embedded, Embodied Interact.* 219–222. doi:10.1145/2460625.2460660

[39] Interhaptics. 2022. Interhaptics Documentation. Retrieved 2024-09-27 from https://www.interhaptics.com/documentation/

[40] Ali Israr and Ivan Poupyrev. 2011. Tactile brush: drawing on skin with a tactile grid display. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '11)*. Association for Computing Machinery, New York, NY, USA, 2019–2028. doi:10.1145/1978942.1979235

[41] Ali Israr, Siyan Zhao, and Oliver Schneider. 2015. Exploring Embedded Haptics for Social Networking and Interactions. In *Proc. ACM Conf. Extended Abstr. Human Factors Comput. Syst.* 1899–1904. doi:10.1145/2702613.2732814

[42] Jaebong Lee, Jonghyun Ryu, and Seungmoon Choi. 2009. Graphical authoring tools for vibrotactile patterns. In *IEEE World Haptics Conf.* 388–389. doi:10.1109/WHC.2009.4810912

[43] Kevin John, Yinan Li, and Hasti Seifi. 2024. AdapTics: A Toolkit for Creative Design and Integration of Real-Time Adaptive Mid-Air Ultrasound Tactons. In *Proc. CHI Conf. Human Factors Comput. Syst.* 1–15. doi:10.1145/3613904.3642090

[44] Markus Jonas. 2008. *Tactile Editor: A Prototyping Tool to Design and Test Vibrotactile Patterns*. Ph. D. Dissertation. RWTH Aachen University.

[45] Yulan Ju, Dingding Zheng, Danny Hynds, George Chernyshov, Kai Kunze, and Kouta Minamizawa. 2021. Haptic Empathy: Conveying Emotional Meaning through Vibrotactile Feedback. In *Extended Abst. CHI Conf. Human Factors Comput. Syst.* 1–7. doi:10.1145/3411763.3451640

[46] Oliver Beren Kaul, Leonard Hansing, and Michael Rohs. 2019. 3DTactileDraw: A Tactile Pattern Design Interface for Complex Arrangements of Actuators. In *Extended Abst. 2019 CHI Conf. Human Factors Comput. Syst.* 1–6. doi:10.1145/3290607.3313030

[47] Y. Kim, J. Cha, J. Ryu, and I. Oakley. 2010. A tactile glove design and authoring system for immersive multimedia. *IEEE MultiMedia* 17, 3 (July 2010), 34–45. doi:10.1109/MMUL.2010.5692181

[48] Youngjae Kim, Heesook Shin, and Minsoo Hahn. 2009. A bidirectional haptic communication framework and an authoring tool for an instant messenger. In *2009 11th Int. Conf. Adv. Commun. Technol.* 2050–2053.

[49] Mathias S Kirkegaard, Mathias Bredholt, Christian Frisson, and Marcelo Wanderley. 2020. TorqueTuner: A self contained module for designing rotary haptic force feedback for digital musical instruments. In *Int. Conf. New Interfaces Musical Expression*. 273–278. doi:10.5281/zenodo.4813359

[50] Khanh-Duy Le, Kening Zhu, Tomasz Kosinski, Morten Fjeld, Maryam Azh, and Shengdong Zhao. 2016. Ubitile: A Finger-Worn I/O Device for Tabletop Vibrotactile Pattern Authoring. In *Proc. Nordic Conf. Human-Comput. Interact.* 1–4. doi:10.1145/2971485.2996721

[51] Hao-Ping (Hank) Lee, Yu-Ju Yang, Thomas Serban Von Davier, Jodi Forlizzi, and Sauvik Das. 2024. Deepfakes, Phrenology, Surveillance, and More! A Taxonomy of AI Privacy Risks. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–19. doi:10.1145/3613904.3642116

[52] Jaebong Lee, J. Ryu, and S. Choi. 2009. Vibrotactile score: A score metaphor for designing vibrotactile patterns. In *World Haptics 2009*. 302–307. doi:10.1109/WHC.2009.4810816

[53] James Leonard and Jerome Villeneuve. 2019. Fast audio-haptic prototyping with mass-interaction physics. Retrieved 2021-08-27 from https://hal.archives-

ouvertes.fr/hal-02015740
[54] James Leonard, Jérôme Villeneuve, and Alexandros Kontogeorgakopoulos. 2020. Multisensory instrumental dynamics as an emergent paradigm for digital musical creation. *Journal on Multimodal User Interfaces* 14, 3 (Sept. 2020), 235–253. doi:10.1007/s12193-020-00334-y
[55] Chungman Lim, Kevin John, Gyungmin Jin, Hasti Seifi, and Gunhyuk Park. 2025. ChatHAP: A Chat-Based Haptic System for Designing Vibrations through Conversation. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–18. doi:10.1145/3706598.3713441
[56] Lofelt. 2022. Lofelt Studio. https://web.archive.org/web/20220526171057/https://lofelt.com/
[57] Maria Luce Lupetti and Dave Murray-Rust. 2024. (Un)making AI Magic: A Design Taxonomy. In *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–21. doi:10.1145/3613904.3641954
[58] Karon E. MacLean, Oliver S. Schneider, and Hasti Seifi. 2017. Multisensory haptic interactions: understanding the sense and designing for it. In *The Handbook of Multimodal-Multisensor Interfaces: Foundations, User Modeling, and Common Modality Combinations*. 97–142.
[59] David Marino, Paul Bucci, Oliver S. Schneider, and Karon E. MacLean. 2017. Voodle: Vocal Doodling to Sketch Affective Robot Motion. In *Proc. Conf. Des. Interact. Syst.* 753–765. doi:10.1145/3064663.3064668
[60] Jonatan Martínez, Arturo S. García, Miguel Oliver, José P. Molina, and Pascual González. 2014. VITAKI: A Vibrotactile Prototyping Toolkit for Virtual Reality and Video Games. *Int. J. Human-Comput. Interact.* 30, 11 (Nov. 2014), 855–871. doi:10.1080/10447318.2014.941272
[61] Francis McGlone and David Reilly. 2010. The cutaneous sensory system. *Neuroscience & Biobehavioral Reviews* 34, 2 (Feb. 2010), 148–159. doi:10.1016/j.neubiorev.2009.08.004
[62] Moritz Alexander Messerschmidt, Sachith Muthukumarana, Nur Al-Huda Hamdan, Adrian Wagner, Haimo Zhang, Jan Borchers, and Suranga Chandima Nanayakkara. 2022. ANISMA: A Prototyping Toolkit to Explore Haptic Skin Deformation Applications Using Shape-Memory Alloys. *ACM Trans. Comput.-Hum. Interact.* 29, 3 (Jan. 2022), 19:1–19:34. doi:10.1145/3490497
[63] Kouta Minamizawa, Yasuaki Kakehi, Masashi Nakatani, Soichiro Mihara, and Susumu Tachi. 2012. TECHTILE toolkit: a prototyping tool for design and education of haptic media. In *Proc. 2012 Virtual Reality Interact. Conf.* 1–2. doi:10.1145/2331714.2331745
[64] Dinmukhammed Mukashev, Nimesha Ranasinghe, and Aditya Shekhar Nittala. 2023. TactTongue: Prototyping ElectroTactile Stimulations on the Tongue. In *Proc. ACM Symp. User Interface Softw. Technol.* 1–14. doi:10.1145/3586183.3606829
[65] Lendy Mulot, Guillaume Gicquel, Quentin Zanini, William Frier, Maud Marchal, Claudio Pacchierotti, and Thomas Howard. 2021. DOLPHIN: A Framework for the Design and Perceptual Evaluation of Ultrasound Mid-Air Haptic Stimuli. In *ACM Symp. Appl. Perception 2021*. 1–10. doi:10.1145/3474451.3476232
[66] S. Nam and D. Fels. 2016. Design and Evaluation of an Authoring Tool and Notation System for Vibrotactile Composition. In *Universal Access Human-Comput. Interact.* 43–53. doi:10.1007/978-3-319-40244-4_5
[67] Neosensory. 2021. Neosensory Developer API. Retrieved 2024-10-03 from https://neosensory.com//wp-content/uploads/2021/04/2021_04_21_-_Neosensory_Developer_API_Documentation-1.pdf
[68] Mathias Nordvall, Mattias Arvola, Emil Boström, Henrik Danielsson, and Timothy Overkamp. 2016. Vibed : A prototyping tool for haptic game interfaces. iSchools, 1–9. Retrieved 2026-01-18 from http://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-126658
[69] Nasrine Olson and Jonas Jarvoll. 2022. Haptic Pattern Designer Toolkit – HaptiDesigner: Software and Hardware for Creation of Actuation Patterns. In *Universal Access Human-Comput. Interact.*, Margherita Antona and Constantine Stephanidis (Eds.). 489–509. doi:10.1007/978-3-031-05028-2_33
[70] M. O'Malley and S. Hughes. 2003. Simplified authoring of 3D haptic content for the World Wide Web. In *11th Symp. Haptic Intact. Virtual Environ. Teleoperator Syst.* 428–429. doi:10.1109/HAPTIC.2003.1191334
[71] Renaud Ott, Vincent De Perrot, Daniel Thalmann, and Frederic Vexo. 2007. MHaptic : a Haptic Manipulation Library for Generic Virtual Environments. In *Int. Conf. Cyberworlds*. 338–345. doi:10.1109/CW.2007.54
[72] Claudio Pacchierotti, Stephen Sinclair, Massimiliano Solazzi, Antonio Frisoli, Vincent Hayward, and Domenico Prattichizzo. 2017. Wearable Haptic Systems for the Fingertip and the Hand: Taxonomy, Review, and Perspectives. *IEEE Transactions on Haptics* 10, 4 (Oct. 2017), 580–600. doi:10.1109/TOH.2017.2689006
[73] Matthew J Page, Joanne E McKenzie, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, Roger Chou, Julie Glanville, Jeremy M Grimshaw, Asbjørn Hróbjartsson, Manoj M Lalu, Tianjing Li, Elizabeth W Loder, Evan Mayo-Wilson, Steve McDonald, Luke A McGuinness, Lesley A Stewart, James Thomas, Andrea C Tricco, Vivian A Welch, Penny Whiting, and David Moher. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* (March 2021), n71. doi:10.1136/bmj.n71
[74] Sabrina Panëels, Margarita Anastassova, and Lucie Brunet. 2013. TactiPEd: Easy Prototyping of Tactile Patterns. In *Human-Comput. Interaction – INTERACT 2013*. 228–245. doi:10.1007/978-3-642-40480-1_15
[75] Sabrina A. Panëels, Panagiotis D. Ritsos, Peter J. Rodgers, and Jonathan C. Roberts. 2013. Prototyping 3D haptic data visualizations. *J. Comput. Graph.* 37, 3 (May 2013), 179–192. doi:10.1016/j.cag.2013.01.009
[76] S. A. Panëels, J. C. Roberts, and P. J. Rodgers. 2010. HITPROTO: a tool for the rapid prototyping of haptic interactions for haptic data visualization. In *2010 IEEE Haptics Symp.* 261–268. doi:10.1109/HAPTIC.2010.5444647
[77] G. Park and S. Choi. 2017. A Physics-Based Vibrotactile Feedback Library for Collision Events. *IEEE Trans. Haptics* 10, 3 (July 2017), 325–337. doi:10.1109/TOH.2016.2614804
[78] E. Pezent, B. Cambio, and M. K. O'Malley. 2021. Syntacts: Open-Source Software and Hardware for Audio-Controlled Haptics. *IEEE Trans. Haptics* 14, 1 (Jan. 2021), 225–233. doi:10.1109/TOH.2020.3002696
[79] Miller S. Puckette. 1996. Pure Data. In *Proceedings of ICMC 1996*. Hong Kong.
[80] J. Ryu and S. Choi. 2008. posVibEditor: Graphical authoring tool of vibrotactile patterns. In *IEEE Int. Workshop Haptic Audio Vis. Environ. Games*. 120–125. doi:10.1109/HAVE.2008.4685310
[81] Oliver Schneider, Karon MacLean, Colin Swindells, and Kellogg Booth. 2017. Haptic experience design: What hapticians do and where they need help. *Int. J. Human-Comput. Stud.* 107 (Nov. 2017), 5–21. doi:10.1016/j.ijhcs.2017.04.004
[82] Oliver Schneider, Siyan Zhao, and Ali Israr. 2015. FeelCraft: User-Crafted Tactile Content. In *Proc. AsiaHaptics*. 253–259. doi:10.1007/978-4-431-55690-9_47
[83] Oliver S. Schneider, Ali Israr, and Karon E. MacLean. 2015. Tactile Animation by Direct Manipulation of Grid Displays. In *Proc. ACM Symp. User Interact. Softw. Technol.* 21–30. doi:10.1145/2807442.2807470
[84] O. S. Schneider and K. E. MacLean. 2014. Improvising design with a Haptic Instrument. In *2014 IEEE Haptics Symp.* 327–332. doi:10.1109/HAPTICS.2014.6775476
[85] O. S. Schneider and K. E. MacLean. 2016. Studying design process and example use with Macaron, a web-based vibrotactile effect editor. In *2016 IEEE Haptics Symp.* 52–58. doi:10.1109/HAPTICS.2016.7463155
[86] Hasti Seifi, Sean Chew, Antony James Nascè, William Edward Lowther, William Frier, and Kasper Hornbæk. 2023. Feellustrator: A Design Tool for Ultrasound Mid-Air Haptics. In *Proc. 2023 CHI Conf. Human Factors Comput. Syst.* 1–16. doi:10.1145/3544548.3580728
[87] H. Seifi, M. Chun, C. Gallacher, O. Schneider, and K. E. MacLean. 2020. How Do Novice Hapticians Design? A Case Study in Creating Haptic Learning Environments. *IEEE Trans. Haptics* 13, 4 (Oct. 2020), 791–805. doi:10.1109/TOH.2020.2968903
[88] Hasti Seifi, Farimah Fazlollahi, Michael Oppermann, John Andrew Sastrillo, Jessica Ip, Ashutosh Agrawal, Gunhyuk Park, Katherine J. Kuchenbecker, and Karon E. MacLean. 2019. Haptipedia: Accelerating Haptic Device Discovery to Support Interaction & Engineering Design. In *Proc. 2019 CHI Conf. Human Factors Comput. Syst.* 1–12. doi:10.1145/3290605.3300788
[89] H. Seifi, K. Zhang, and K. E. MacLean. 2015. VibViz: Organizing, visualizing and navigating vibration libraries. In *IEEE World Haptics Conf.* 254–259. doi:10.1109/WHC.2015.7177722
[90] Stephen Sinclair and Marcelo M. Wanderley. 2007. Defining A Control Standard For Easily Integrating Haptic Virtual Environments With Existing Audio / Visual Systems. In *Proc. Conf. New Interfaces Musical Expression.* doi:10.5281/ZENODO.1177245
[91] Stephen Sinclair and Marcelo M. Wanderley. 2009. A run-time programmable simulator to enable multi-modal interaction with rigid-body systems. *J. Interact. Comput.* 21, 1-2 (Jan. 2009), 54–63. doi:10.1016/j.intcom.2008.10.012
[92] Tanay Singhal and Oliver Schneider. 2021. Juicy Haptic Design: Vibrotactile Embellishments Can Improve Player Experience in Games. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3411764.3445463
[93] Paul Strohmeier, Seref Güngör, Luis Herres, Dennis Gudea, Bruno Fruchard, and Jürgen Steimle. 2020. bARefoot: Generating Virtual Materials using Motion Coupled Vibration in Shoes. In *Proc. ACM Symp. User Interface Softw. Technol.* 579–593. doi:10.1145/3379337.3415828
[94] Youjin Sung, Kevin John, Sang Ho Yoon, and Hasti Seifi. 2025. HapticGen: Generative Text-to-Vibration Model for Streamlining Haptic Design. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, 1–24. doi:10.1145/3706598.3713609
[95] Youjin Sung, Rachel Kim, Kun Woo Song, Yitian Shao, and Sang Ho Yoon. 2023. HapticPilot: Authoring In-situ Hand Posture-Adaptive Vibrotactile Feedback for Virtual Reality. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 4 (Dec. 2023), 1–28. doi:10.1145/3631453
[96] C. Swindells, E. Maksakov, K. E. MacLean, and V. Chung. 2006. The Role of Prototyping Tools for Haptic Behavior Design. In *14th Symp. Haptic Interact. Virtual Environ. Teleoperator Syst.* 161–168. doi:10.1109/HAPTIC.2006.1627084

[97] C. Swindells, S. Pietarinen, and A. Viitanen. 2014. Medium fidelity rapid prototyping of vibrotactile haptic, audio and video effects. In *2014 IEEE Haptics Symp.* 515–521. doi:10.1109/HAPTICS.2014.6775509
[98] Janet L. Taylor. 2022. Kinesthetic Inputs. In *Neuroscience in the 21st Century*, Donald W. Pfaff, Nora D. Volkow, and John L. Rubenstein (Eds.). Springer International Publishing, Cham, 1339–1373. doi:10.1007/978-3-030-88832-9_31
[99] Yusuke Ujitoko, Yuki Ban, and Koichi Hirota. 2020. GAN-Based Fine-Tuning of Vibrotactile Signals to Render Material Surfaces. *IEEE Access* 8 (2020), 16656–16661. doi:10.1109/ACCESS.2020.2968185
[100] Anke van Oosterhout, Miguel Bruns, and Eve Hoggan. 2020. Facilitating Flexible Force Feedback Design with Feelix. In *Proc. 2020 Int. Conf. Multimodal Interact.* 184–193. doi:10.1145/3382507.3418819
[101] Anke van Oosterhout, Miguel Bruns, and Eve Hoggan. 2023. Designing for Haptic and Embodied Interaction with Feelix. In *Proc. Int. Conf. Tangible, Embedded, Embodied Interact.* 1–4. doi:10.1145/3569009.3571842
[102] Steeven Villa and Sven Mayer. 2022. Cobity: A Plug-And-Play Toolbox to Deliver Haptics in Virtual Reality. In *Proc. Mensch und Computer 2022.* 78–84. doi:10.1145/3543758.3543775
[103] Preeti Vyas, Unma Mayur Desai, Karin Yamakawa, and Karon Maclean. 2023. A Descriptive Analysis of a Formative Decade of Research in Affective Haptic System Design. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23).* Association for Computing Machinery, New York, NY, USA, 1–23. doi:10.1145/3544548.3580735
[104] Anna Weisling, Anna Xambó, Ireti Olowe, and Mathieu Barthet. 2018. Surveying The Compositional And Performance Practices Of Audiovisual Practitioners. In *Proceedings of the International Conference on New Interfaces for Musical Expression.* Blacksburg, Virginia, USA, 344–345. doi:10.5281/ZENODO.1302608
[105] Dennis Wittchen, Bruno Fruchard, Paul Strohmeier, and Georg Freitag. 2021. TactJam: a collaborative playground for composing spatial tactons. In *Proc. Int. Conf. Tangible, Embedded, Embodied Interact.* 1–4.
[106] Dennis Wittchen, Katta Spiel, Bruno Fruchard, Donald Degraen, Oliver Schneider, Georg Freitag, and Paul Strohmeier. 2022. TactJam: An End-to-End Prototyping Suite for Collaborative Design of On-Body Vibrotactile Feedback. In *Proc. Int. Conf. Tangible, Embedded, Embodied Interact.* 1–13. doi:10.1145/3490149.3501307
[107] Jacob O. Wobbrock and Julie A. Kientz. 2016. Research contributions in human-computer interaction. *interactions* 23, 3 (April 2016), 38–44. doi:10.1145/2907069
[108] Takehiko Yamaguchi, Kazuhiro Oshima, Yuto Hirano, Akiyo Makishima, Tetsuya Harada, and Paul Richard. 2015. Component-based Authoring Tool for Haptic Navigation:. In *Proc. Int. Conf. Comput. Graph. Theory Appl.* 486–491. doi:10.5220/0005357804860491
[109] Siyan Zhao, Oliver Schneider, Roberta Klatzky, Jill Lehman, and Ali Israr. 2014. FeelCraft: crafting tactile experiences for media using a feel effect library. In *Proc. Adjunct ACM Symp. User Interact. Softw. Technol.* 51–52. doi:10.1145/2658779.2659109

A Taxonomy Examples

The selected haptic authoring tools described here are intended to aid the reader in understanding how the current taxonomy (Section 6) can be applied to real projects. Readers interested in reviewing the 79 collected tools should consult the list in the paper's supplemental materials or the Haptic Authoring Toolbox website at https://srl.mcgill.ca/hat-box.

A.1 Macaron

Macaron [85] (Table 5, first column) is a web-based (Platform) vibrotactile authoring tool (Category) created to research how people perform haptic design (Use Case). Sound synthesis within the tool can be used to drive any voice-coil actuators (Abstraction, Device Names). No functionality is present to aid the user in supporting other types of hardware devices (Device Template) or to design for a specific part of the body (Body Positions).

The interface features editing tracks where the user can add keyframes (UI Metaphors) to construct a time-based pattern (Driving Feature). Each keyframe specifies the intensity and frequency of vibration at that point, with in-between values linearly interpolated. A library of example patterns are included within the tool (Design Approaches). As the user edits, they can play back the effect (Iterative Playback) directly on a connected device (Effect Localization). The user can export the pattern as a JSON file for future editing or as a WAV file for deployment (Storage). No functionality is present to work with other sensory modalities (Non-Haptic Media) or to link the tool to other software (Connectivity).

Macaron was published for the first time at the 2016 (Release Year) IEEE Haptics Symposium (Venues) and can be accessed via a public website[8] (Availability). Despite its source code being available in a Git repository,[9] no rights are explicitly granted or reserved (License).

A.2 Cobity

Cobity [102] (Table 5, second column) is an authoring tool built in Unity (Platform) to create room-scale force-feedback effects (Category) for virtual reality (Use Case). To accomplish this, the tool controls a Kinova Gen3 robot (Device Names, Abstraction) to position objects around someone in a VR experience. Cobity tracks this person's movement so that they feel the end effector of the robot when touching a virtual object (Driving Feature, Effect Localization). No explicit functionality exists to extend support to other devices (Device Template) or to design on a specific part of the body (Body Positions).

Within Unity, Cobity is accessed through traditional GUI elements such as input fields and dropdown menus (UI Metaphors). Using this interface, a designer can set where the physical location of the end effector or have it track the position of a virtual object (Design Approaches). Designers can test what they create with the tool by playing the scene in Unity (Iterative Playback). Despite being a plug-in to a game engine, visual and audio elements are not present within Cobity's interface (Non-Haptic Media). No functionality is specified for saving effects (Storage) or connecting the tool to other software than Unity (Connectivity).

Cobity was published for the first time in 2022 (Release Year) at the ACM Mensch und Computer conference (Venues). It can be downloaded from the Git repository[10] of the project (Availability) and is open-source (License) under the terms of the MIT software license.

B Attribute Summary Tables

Tables 6, 7, and 8 show the most common values for each attribute in the 79-tool collection. Each cell begins with the name of the value followed by a reference to a tool that exemplifies it. Below them, the total number and percentage of tools with this value are shown. Attributes followed by an asterisk have more values in the collection than those shown in the table.

[8] https://hapticdesign.github.io/macaron/
[9] https://github.com/hapticdesign/hapticdesign.github.io
[10] https://github.com/xteeven/Cobity

Table 5: Application of the current taxonomy to selected tools

	Macaron	**Cobity**
Release Year	2016	2022
Availability	Available	Available
License	Unknown	Open Source
Use Case	Tool research	Virtual reality
Platform	Web	Unity
Venues	Haptics Symposium	Mensch und Computer
Category	Vibrotactile	Force Feedback
Device Names	Voice Coil	Kinova Gen3
Abstraction	Class	Consumer
Device Template	No	No
Body Positions	N/A	N/A
Driving Feature	Time	Action
Effect Localization	Device-centric	Target-centric
Iterative Playback	Yes	Yes
Non-Haptic Media	None	None
Design Approaches	Direct, Procedural, Library	Direct, Procedural
UI Metaphors	Track, Keyframe	None
Storage	Custom JSON, WAV	None
Connectivity	None	None

Table 6: Metadata taxonomy attributes and examples.

Metadata Attribute	**Examples**				
Release Year*	From: 1995 [10]	To: 2024 [43]			
Availability	Unavailable [74] 40 (51%)	Available [78] 39 (49%)			
License	Unknown [85] 44 (56%)	Open Source [101] 27 (34%)	Proprietary [3] 7 (8.9%)	Non-Commercial [19] 1 (1.3%)	
Use Case*	Tool research [84] 21 (27%)	Hardware [93] 15 (19%)	Augmentation [47] 15 (19%)	Virtual Reality [95] 13 (16%)	Simulation [53] 11 (14%)
Platform*	Windows [69] 26 (33%)	macOS [44] 12 (14%)	Linux [90] 11 (14%)	Unknown [46] 9 (11%)	Unity [102] 8 (10%)
Venues*	Haptics Symp. [97] 8 (10%)	CHI [40] 6 (7.6%)	UIST [64] 6 (7.6%)	World Haptics [12] 4 (5.1%)	NIME [31] 3 (3.8%)

Table 7: Hardware taxonomy attributes and examples.

Hardware Attribute	Examples				
Category	Vibrotactile [46] 54 (68%)	Force feedback [28] 26 (33%)	Temperature [48] 3 (3.8%)	Skin stretch [62] 2 (2.5%)	Electrotactile [64] 1 (1.3%)
Device Names*	Voice Coil [78] 11 (14%)	Phantom [19] 8 (10%)	Falcon [5] 7 (8.9%)	Android [41] 6 (7.6%)	iPhone [68] 5 (6.3%)
Abstraction	Consumer [86] 37 (47%)	Custom [105] 22 (28%)	Class [85] 19 (24%)		
Device Template	No [37] 74 (94%)	Yes [16] 5 (6.3%)			
Body Positions*	N/A [85] 54 (68%)	Hand [43] 11 (14%)	Arm [29] 10 (13%)	Torso [69] 8 (10%)	Head [46] 5 (6.3%)

Table 8: Interaction taxonomy attributes and examples.

Interaction Attribute	Examples				
Driving Feature	Time [66] 52 (66%)	Action [96] 38 (48%)			
Effect Localization	Device-centric [48] 35 (44%)	Location-aware [60] 25 (32%)	Target-centric [95] 19 (24%)		
Iterative Playback	Yes [85] 61 (77%)	N/A [36] 15 (19%)	No [38] 3 (3.8%)		
Non-Haptic Media	None [99] 51 (65%)	Audio [34] 21 (27%)	Visual [91] 19 (24%)	Olfactory [15] 1 (1.3%)	
Design Approaches	Direct [75] 61 (77%)	Procedural [83] 57 (72%)	Additive [80] 27 (34%)	Library [97] 24 (30%)	Description [89] 5 (6.3%)
UI Metaphors	Demonstration [59] 27 (34%)	Track [23] 24 (30%)	Keyframe [68] 22 (28%)	Dataflow [5] 6 (7.6%)	Music Notation [52] 2 (2.5%)
Storage*	None [59] 24 (30%)	Unknown [37] 13 (16%)	XML [80] 10 (13%)	JSON [85] 9 (11%)	WAV [63] 8 (10%)
Connectivity	None [50] 45 (57%)	API [78] 22 (28%)	OSC [49] 5 (6.3%)	Other [43] 3 (3.8%)	